Machine Learning and Neural Networks

DeepPumas Workshop

Authors: Andreu Vall, Mohamed Tarek, and Niklas Korsbo

Pumas-AI Inc

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Broad types of machine learning

- Supervised machine learning
  - Prediction
  - Classification
- Unsupervised machine learning
  - Generative models
  - Feature extraction
  - Clustering
- Semi-supervised machine learning
  - Has components of both supervised and unsupervised learning
  - Examples:
    - Semi-supervised conditional generative models
    - Feature extraction followed by supervised ML

For depth here,
check out our AIDD course.

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Fundamental Concepts of Machine Learning and Neural Networks

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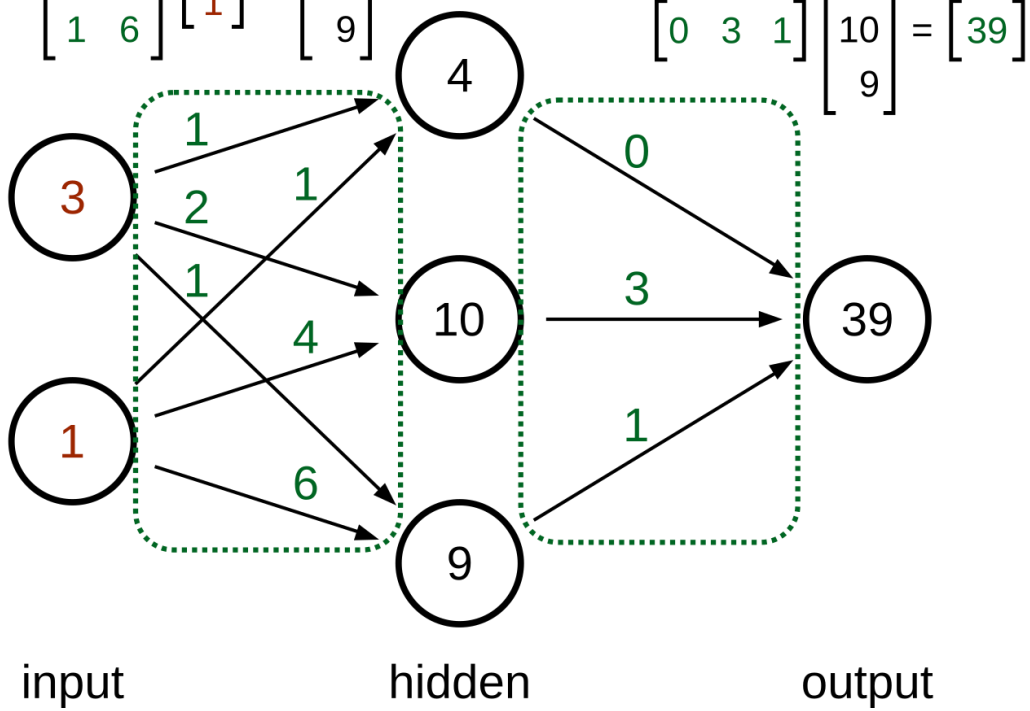
Neural Networks

f = g = identity

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ 9 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 10 \\ 9 \end{bmatrix} = \begin{bmatrix} 39 \end{bmatrix}$$



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Training

- Find network parameters $\mathbf{W}$ that minimize a suitable cost function

$$R_{\text{emp}}\left(\mathbf{W}, \mathbf{X}, \mathbf{Y}\right) = \frac{1}{N} \sum_{n=1}^{N} L\left(y^n, \hat{y}^n(\mathbf{x}^n; \mathbf{W})\right)$$

- Iteratively refine the parameters by gradient-based optimization

$$\mathbf{W}^{\text{new}} = \mathbf{W}^{\text{old}} - \eta \nabla_{\mathbf{W}} R_{\text{emp}}\left(\mathbf{W}, \mathbf{X}, \mathbf{Y}\right)$$

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Control on Withheld Data

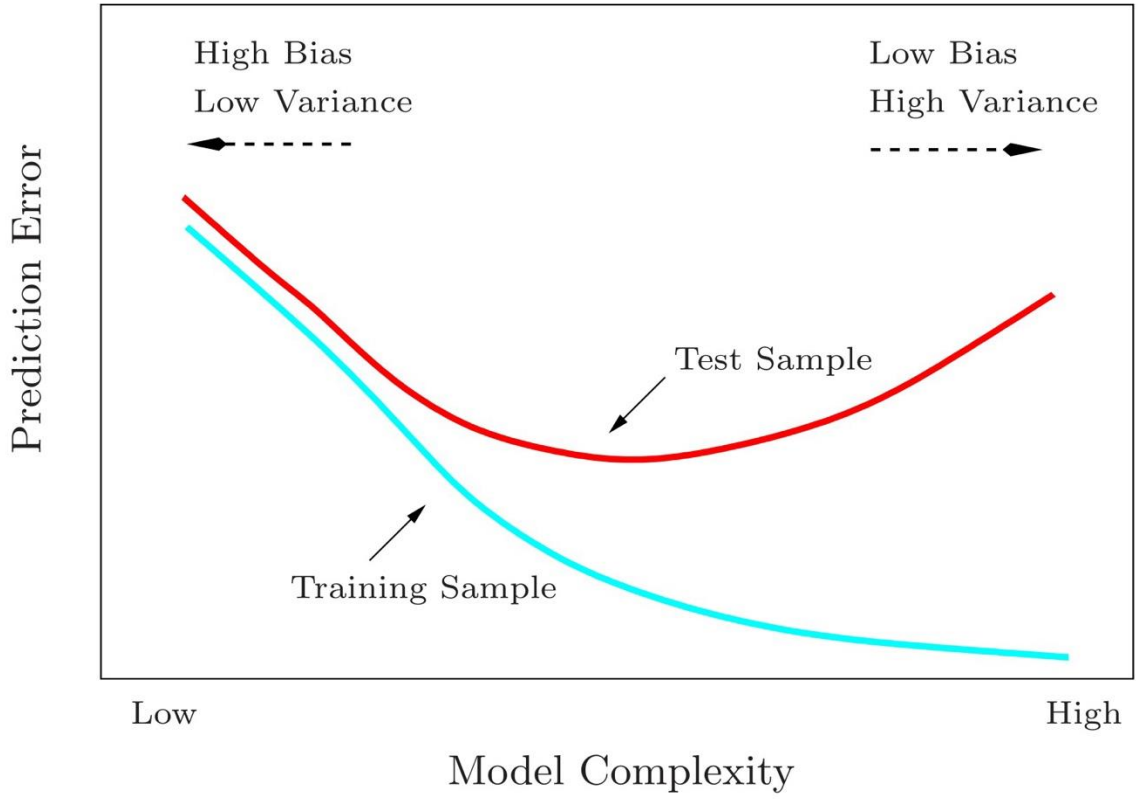

FIGURE 2.11. *Test and training error as a function of model complexity.*

Figure by Hastie et al., 2008

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Practical Guide to Training Neural Networks

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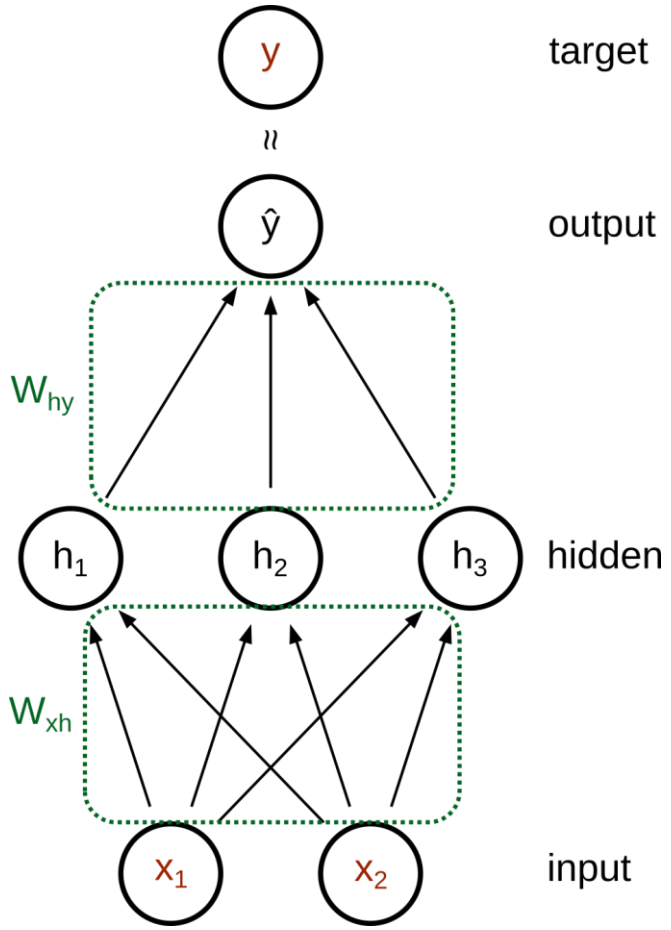
Feed-Forward Networks

- The basic interface is quite given by the task
  - Number of input units
  - Number of output units
  - Activation function for the output layer

- The *architecture* must be selected
  - Number of hidden layers
  - Number of units per hidden layer
  - Activation function for the hidden layers
  - Parameters of the optimization algorithm



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Activation Functions

- At the output layer we typically use
  - the identity function for regression tasks,
  - the sigmoid function for binary classification tasks,
  - the softmax function for multiclass classification tasks, and
  - the softplus function to ensure non-negativity.

- At the hidden layers,
  - traditional activations are the sigmoid and tanh functions,
  - but deeper networks will suffer from vanishing gradients.
  - In this case, experiment with the ReLU and SELU functions.

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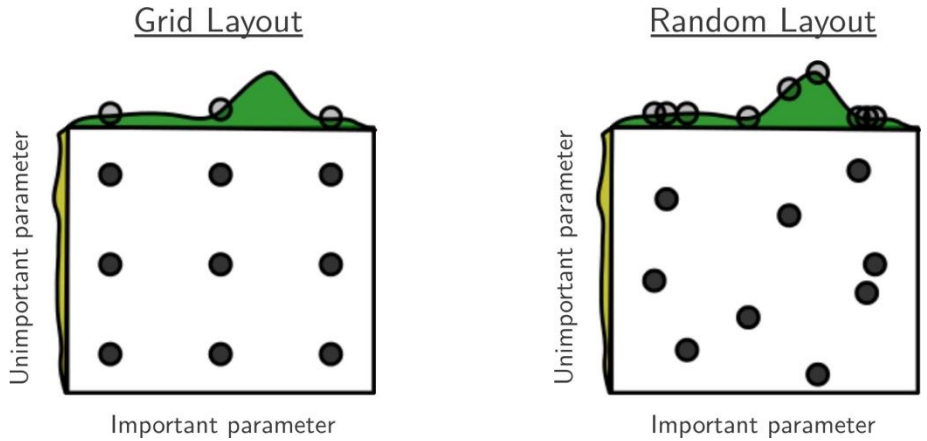
Architecture Selection

- Typically based on empirical investigation and assessment on withheld data
- Initial exploration to get a feel of reasonable network dimensions
- Hyperparameter search
  - Design a grid of hyperparameters based on the previous exploration
  - In case of a tight budget, pay attention to the learning rate
  - Exhaustive, random, Bayesian search



Bergstra and Bengio "Random search for hyper-parameter optimization." (JMLR, 2012)

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Underfitting and Overfitting

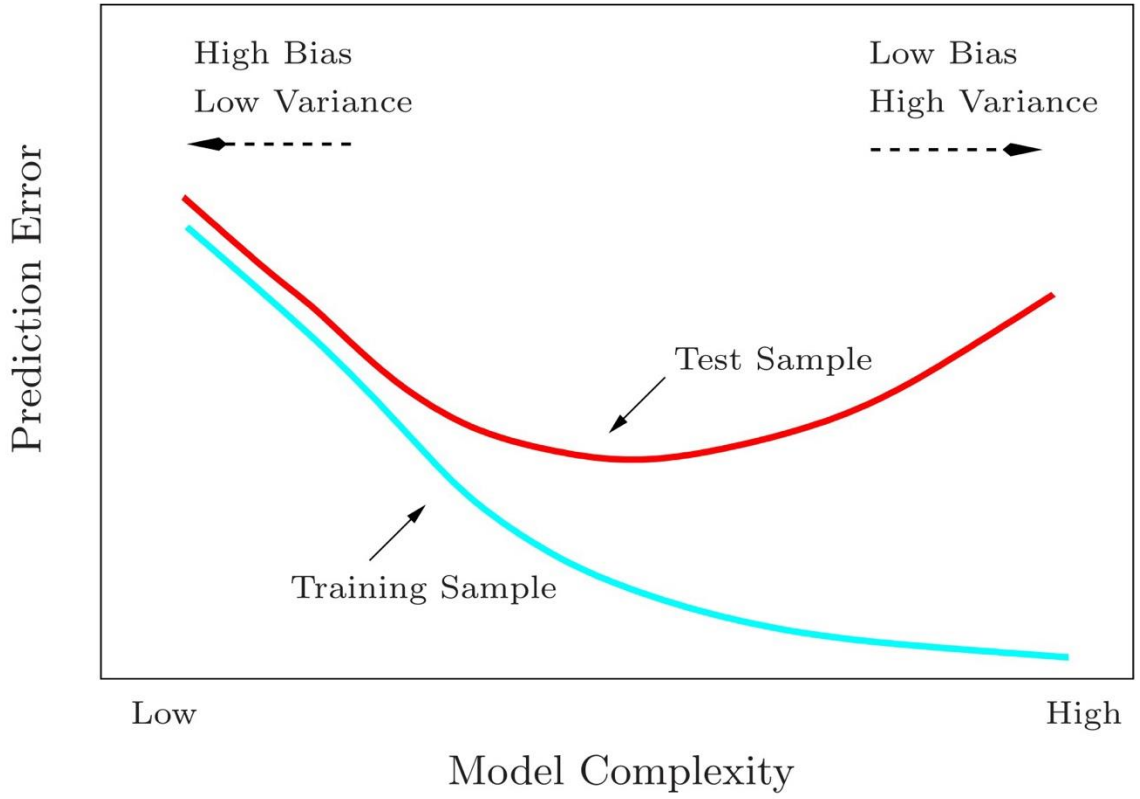

FIGURE 2.11. *Test and training error as a function of model complexity.*

Figure by Hastie et al., 2008

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Strategies to Prevent Overfitting

- Early stopping based on withheld data
- Most careful data splitting to avoid data leakage
- Networks as large as possible, but not larger
- Larger and more diverse training datasets
- Regularization
  - Penalization of the cost function
  - Dropout (Srivastava et al., 2014)
  - Data *augmentation*

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Penalization of the Cost Function

- Cost function

$$\arg\min_{\mathbf{W}} \frac{1}{N} \sum_{n=1}^{N} L\left(y^n, \hat{y}^n(\mathbf{x}^n; \mathbf{W})\right)$$

- Penalized cost function reduces expressiveness

$$\arg\min_{\mathbf{W}} \frac{1}{N} \sum_{n=1}^{N} L\left(y^n, \hat{y}^n(\mathbf{x}^n; \mathbf{W})\right) + \lambda \|\mathbf{W}\|_p$$

- Typically using $L1$ or $L2$ norm

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Hands-on session: Fitting, Overfitting and Regularizing Neural Networks

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Conditional generative models

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Generative models

- Definitions
  - $\boldsymbol{z}$: latent variables of dimension d
  - $\boldsymbol{y}$: observed response/data
  - $\boldsymbol{y_g}$: generated/simulated/synthetic response/data
- Model

$$\boldsymbol{y_g} = \boldsymbol{f}(\boldsymbol{z}) + \boldsymbol{\epsilon}$$
$$\boldsymbol{z} \sim \text{Normal}(0, \boldsymbol{I}_{d \times d})$$
$$\epsilon_i \sim \text{Normal}(0, \sigma^2)$$

- Objective: choose $\boldsymbol{f}$ such that the distribution of $\boldsymbol{y_g}$ is close to the distribution of the observed data $\boldsymbol{y}$

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Conditional generative models

- Definitions
  - $\boldsymbol{z}$: latent variables of dimension d
  - $\boldsymbol{x}$: observed covariates
  - $\boldsymbol{y}$: observed response
  - $\boldsymbol{y_g}$: generated/simulated/synthetic response
- Model

$$\boldsymbol{y_g} = \boldsymbol{f}(\boldsymbol{z}, \boldsymbol{x}) + \boldsymbol{\epsilon}$$
$$\boldsymbol{z} \sim \text{Normal}(0, \boldsymbol{I}_{d \times d})$$
$$\epsilon_i \sim \text{Normal}(0, \sigma^2)$$

- Objective: choose $\boldsymbol{f}$ such that the conditional distribution of $\boldsymbol{y_g} \mid \boldsymbol{x}$ is close to the conditional distribution of the observed data $\boldsymbol{y} \mid \boldsymbol{x}$

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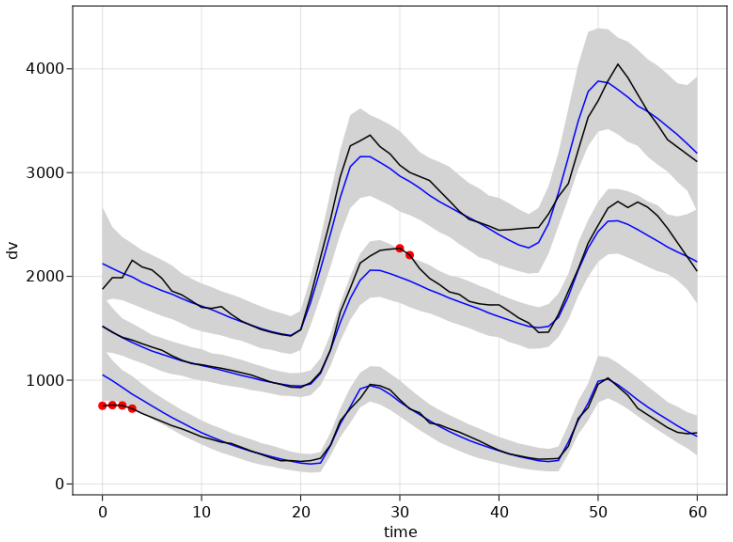
Population PK is machine learning!

- Definitions
  - $\boldsymbol{\eta}$: latent variables of dimension d and covariance matrix $\boldsymbol{\Omega}$
  - $\boldsymbol{x}$: observed covariates
  - $\mathbf{dv}$: observed response
  - $\mathbf{dv_g}$: generated/simulated/synthetic response
- Model

$$\mathbf{dv_g} = \boldsymbol{f_\theta}(\boldsymbol{\eta}, \boldsymbol{x}) + \boldsymbol{\epsilon}$$
$$\boldsymbol{\eta} \sim \text{Normal}(0, \boldsymbol{\Omega})$$
$$\epsilon_i \sim \text{Normal}(0, \sigma^2)$$



- Objective: choose $\boldsymbol{f_\theta}$ such that the conditional distribution of $\mathbf{dv_g} \mid \boldsymbol{x}$ is close to the conditional distribution of the observed data $\mathbf{dv} \mid \boldsymbol{x}$

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